Team Information

Project Title: Elementopia

Project Short Description: Elementopia is an interactive, web-based gamified learning platform that transforms complex chemistry concepts into engaging adventures for high school students.

Team Code: 2526-sem2-it332-53

Members:

1. Ocampo, Danielle Maxine P.
2. Benolirao, Clyde C.
3. Ceniza, Jian Marc
4. Dabalos, Erica Y.
5. Dibdib, Wayne Kenji B.

PART 1: Introduction

Chemistry remains one of the most challenging subjects for high school students due to its abstract concepts and minimal engagement in traditional learning environments. Students frequently struggle because core concepts such as chemical bonding are notoriously difficult to visualize in a standard classroom. A survey of 25 high school students confirmed that 72% find it hard to memorize chemical elements, and 96% expressed a preference for learning within a gamified environment. Research supports this direction, a structurally coherent gamification promotes cognitive processing and shifts students from passive memorization toward meaningful, goal-directed tasks (Kouneli et al., 2024; Antonaci et al., 2023).

This project builds upon an existing system, Elementopia, originally developed as a web-based gamified chemistry platform featuring simulated bonding mechanics, a sandbox laboratory, and teacher-managed content. While the original system demonstrated the viability of gamification in chemistry education, it relied primarily on quiz-based assessments and binary right-or-wrong feedback, limiting its capacity for discovery-oriented learning. It also lacked any mechanism to measure how students reason through problems, only whether they reached a correct answer.

The formal problem this project addresses is: existing gamified chemistry platforms assess what students know through quiz outcomes rather than how students reason through problems, producing no measurable data on problem-solving behavior or error recovery.

To address this gap, this project extends Elementopia by introducing Elemental Resonance Domains as a new core learning mechanic — text-free environmental puzzle rooms where students overcome physical obstacles by synthesizing chemical compounds to trigger environmental reactions that clear their path forward. Instead of answering text questions, students manipulate their environment, and the system tracks compound accuracy and completion efficiency as performance indicators. This approach is supported by evidence that discovery-based, simulation-driven learning significantly improves conceptual retention over quiz-based formats (Li et al., 2025; Angell et al., 2024).

PART 2: Objectives

This section defines the intended outcomes of the proposed system, mapped directly to major functional modules. The objectives are outcome-based, shifting the educational focus from rote memorization to the development of algorithmic thinking.

A. General Objectives

- **General Objective 1:** Improve applied chemistry problem-solving skills to an 80% Domain success rate by automating an Elemental Resonance Domain puzzle system that eliminates text-based assessments and reduces random guessing through Domain puzzle rooms where students synthesize compounds to trigger environmental reactions that clear physical obstacles blocking their progression.
- **General Objective 2:** Automate session-based learner performance tracking via Supabase to improve self-directed knowledge gap identification efficiency by 90% through a personalized visual Mastery Dashboard without requiring user registration.
- **General Objective 3:** Optimize peer-to-peer competitive multiplayer matchmaking to reduce connection time to under 5 seconds and enhance algorithmic problem-solving application under pressure through real-time Speed-to-Compound evaluation with 100% computational accuracy.

B. Specific Objectives

Module 1: Elemental Resonance Puzzle Module

- Improve problem-solving accuracy to 80% by replacing text-based assessments with Elemental Resonance Domain puzzle rooms where students synthesize chemical compounds to trigger environmental reactions that clear physical obstacles blocking their progression.
- Improve error recovery by generating Meaningful Byproduct diagnostics for a predefined set of incorrect compound combinations, replacing binary "Wrong" notifications with contextual chemistry-based feedback.
- Reduce random guessing by automating a Hazmat Protocol failsafe that disables irrelevant elements upon detecting 5 failed attempts within 15 seconds.
- Enhance logical progression by requiring a minimum sequence of 3 correct reactions per Domain before unlocking the next room.

- Improve learning curve structure by restricting access to Compound Chambers until foundational Element Rooms are cleared.
- Reduce repetitive memorization by automating randomized compound selection from a predefined pool per session.
- Increase sustained session engagement to greater than 20 minutes through completion banners.

Module 2: Assessment and Progress Module

- Automate session performance logging (compound accuracy and completion speed) into Supabase within less than 2 seconds per session.
- Improve knowledge gap identification by generating a visual Mastery Dashboard that enables 90% of students to independently target weak Domains without teacher intervention.
- Improve overall proficiency accuracy to 100% by consolidating match results from the Opponent-Based Challenge Module into the session tracker.

Module 3: Opponent-Based Challenge Module

- Reduce matchmaking connection time to under 5 seconds by implementing auto-generated 5-digit access codes and temporary session nicknames.
- Ensure 100% unique competitive puzzle scenarios per match by automating randomized compound task selection from a predefined pool without manual input.
- Improve match result accuracy to 100% by automating Speed-to-Compound evaluation logic that determines winners based strictly on reaction path efficiency via Supabase Realtime.

Research Questions

1. To what extent does the Elemental Resonance Domain system improve high school students' applied chemistry problem-solving skills in terms of Domain success rate, reduction of random guessing through the Hazmat Protocol, and error recovery through Meaningful Byproduct diagnostics?
2. How effective is the Assessment and Progress Module in improving self-directed learning efficiency in terms of performance tracking speed, accuracy of proficiency calculation, and the student's ability to independently identify knowledge gaps through the Mastery Dashboard?
3. How does the Opponent-Based Challenge Module improve algorithmic problem-solving under pressure in terms of matchmaking connection time, uniqueness of competitive scenarios, and accuracy of Speed-to-Compound winner determination?

PART 3: Methods

Elementopia is an interactive, web-based gamified chemistry education platform designed to improve high school students' applied chemistry problem-solving skills and algorithmic thinking. This project extends an existing system inherited from a prior development team by introducing three new modules built on top of the original platform's features: the Elemental Resonance Puzzle Module, the Assessment and Progress Module, and the Opponent-Based Challenge Module. The platform targets high school students who struggle with traditional text-heavy chemistry instruction. It is developed using React for the frontend, Spring Boot for backend logic and API management, and Supabase (PostgreSQL) for persistent session data storage and real-time multiplayer synchronization.

The project follows an Agile development methodology to support structured, iterative development within a constrained timeframe. Development is divided into four sprints. Sprint 1 focuses on the core Elemental Resonance Domain mechanic, including environmental obstacles, compound validation logic, Meaningful Byproduct diagnostics, and the Hazmat Protocol failsafe. Sprint 2 covers the Assessment and Progress Module, specifically Supabase session logging and the visual Mastery Dashboard. Sprint 3 integrates the Opponent-Based Challenge Module, implementing 5-digit access code matchmaking and Supabase Realtime-powered Speed-to-Compound evaluation. Sprint 4 is dedicated to system-wide testing, mobile responsiveness optimization, and final integration.

The platform will be validated using a quantitative approach. Performance effectiveness will be measured through automated telemetry, targeting an 80% Domain success rate and 100% computational accuracy in match winner determination. System performance will be assessed by tracking response times, targeting under 2 seconds for session data logging and under 5 seconds for multiplayer matchmaking connection. User engagement and acceptance will be evaluated through post-session surveys targeting a 90% self-directed knowledge gap identification rate, demonstrating the system's effectiveness as a self-directed learning tool.

PART 4: Expected System

The anticipated outcome of this project is three new functional modules integrated into the existing Elementopia platform. The Minimum Viable Product focuses on the core deliverable of each module at a minimum of 60% functional completion. The Elemental Resonance Puzzle Module delivers interactive Domain puzzle rooms where students synthesize chemical compounds to trigger environmental reactions that clear physical obstacles blocking their progression, with Meaningful Byproduct diagnostics generated for a predefined set of incorrect combinations to guide conceptual understanding.. The Assessment and Progress Module delivers automated session performance logging to Supabase and a visual Mastery Dashboard that displays compound accuracy and completion speed per Domain. The Opponent-Based Challenge Module delivers functional room creation and joining via auto-generated 5-digit access codes, with both players presented the same compound puzzle simultaneously through Supabase Realtime.

The system workflow begins with the student entering a temporary session nickname, bypassing traditional authentication. The system retrieves the student's existing performance data from Supabase and renders their Mastery Dashboard. From this interface, the student selects either a self-paced Elemental Resonance Domain or enters a 5-digit code to join a competitive match. During active gameplay, the student interacts with the Domain environment by combining chemical elements to synthesize the correct compound. Invalid combinations trigger Meaningful Byproduct diagnostics displayed directly on the interface. Upon successful compound synthesis, the environmental obstacle is cleared, the system logs completion speed and compound accuracy to Supabase, and the Mastery Dashboard is updated accordingly.

PART 5: Discussion

Scope of the System

The proposed extension of Elementopia is designed specifically for high school students to develop applied chemistry knowledge and algorithmic thinking through interactive visual problem-solving. The functional scope is strictly bounded to three new modules built on top of the existing platform: the Elemental Resonance Puzzle Module for Domain puzzle rooms where students synthesize chemical compounds to trigger environmental reactions that clear physical obstacles; the Assessment and Progress Module for automated, device-agnostic session tracking via Supabase; and the Opponent-Based Challenge Module for real-time peer-to-peer matchmaking via auto-generated 5-digit access codes. The platform maintains the original system's full mobile responsiveness as a core architectural requirement, ensuring equitable access and seamless interaction across personal devices without requiring user registration.

Limitations of the Project

Several constraints affect the project's development and operational deployment. First, due to a strict development timeframe, the chemical compound pool is restricted to a predefined set of high school-appropriate reactions rather than an exhaustive simulation of all periodic table interactions. Second, as a deliberate technical constraint to maintain a frictionless onboarding experience, the system does not implement user registration or authentication. All session data is persisted to Supabase (PostgreSQL), meaning performance history is device-agnostic and accessible across sessions as long as the student uses the same temporary nickname. However, if a student changes their nickname, prior session data will not be associated with their new identity. Third, session data persistence is tied to the student's temporary nickname — clearing the nickname or switching devices without re-entering the same nickname will result in a disconnected performance history. Finally, the platform intentionally omits teacher administration tools, restricting evaluation to immediate session-based performance metrics rather than long-term academic monitoring.

Expected Contribution

This project directly addresses the research gap identified in Part 1 — that existing gamified chemistry platforms, including the original Elementopia, measure only quiz outcomes rather than how students reason through problems. By introducing Elemental Resonance Domains, the system shifts assessment from answer correctness to problem-solving behavior, capturing compound accuracy and completion efficiency as meaningful performance indicators. Furthermore, the Opponent-Based Challenge Module introduces competitive pressure as a driver of algorithmic thinking, exposing students to real-time decision-making under time constraints. Collectively, these three modules transform Elementopia from a gamified quiz platform into a discovery-driven, behaviorally-tracked learning environment that empowers students to autonomously identify and address their own conceptual gaps.

PART 6: Traceability Matrix

The following matrix ensures alignment between related research evidence and the simplified system features designed for high school students:

RRL Finding/Theme	Identified Gap	Research Question	Proposed Function
Interactive Gamification in Chemistry Education: Gamification serves as a pivotal instrument in chemistry-related education. It fosters active student engagement, enhances the practical application of knowledge, and allows students to experiment in a safe, interactive environment (*The Infusion of Gamification in Promoting Chemical Engineering Laboratory Classes*, 2023). Additionally, structurally coherent gamification promotes cognitive processing, shifting students away from rote memorization toward meaningful, goal-directed tasks (*Potions & Dragons*, MDPI, 2025).	Current digital simulations often lack structured error-recovery mechanisms and failsafes against brute-force guessing. They frequently rely on binary "right/wrong" feedback instead of providing diagnostic guidance when students make conceptual mistakes.	To what extent does the Interactive Puzzle Module autonomously improve high school students' applied chemistry knowledge in terms of sequential problem-solving success, reduction of random guessing, and application of error recovery logic?	**Elemental Resonance Puzzle Module** featuring Domain puzzle rooms where compound synthesis triggers environmental reactions, supported by Meaningful Byproduct diagnostics and a Hazmat Protocol failsafe.

Learning Analytics & Self-Regulated Learning: Learning Analytics Dashboards (LADs) are critical tools designed to foster self-regulated learning (SRL) by visualizing performance and progress. Studies demonstrate that dashboards combining clear visualizations with actionable feedback significantly increase learners' behavioral commitment, helping them reflect on their behavior and adapt their learning strategies autonomously (*Design Principles and Impact of a Learning Analytics Dashboard*, MDPI, 2025).	Standard educational tools and dashboards often base progress on flat, binary quiz scores rather than calculating real problem-solving efficiency, reaction paths, and logical progression across varying difficulty levels.	How effective are the Learning Rooms and the Self-paced Efficiency Assessment Module in facilitating independent conceptual understanding in terms of logical progression, calculating efficiency, and achieving learner satisfaction?	**Assessment and Progress Module** featuring automated session performance logging to Supabase and a visual Mastery Dashboard that enables students to independently identify knowledge gaps without teacher intervention.
Competitive Gamification under Time Pressure: Introducing competitive structures into technology-mediated training promotes flow and engagement. Research indicates that matching learners in opponent-based scenarios directly impacts their self-efficacy and real-time application of knowledge, though the design must thoughtfully balance conceptual learning outcomes with	Few chemistry-specific platforms autonomously generate unique, randomized puzzles that evaluate real-time reaction path efficiency against an opponent, instead relying on static, text-based competitive trivia.	How does the integration of the Random Compound Challenge System and Opponent-Based Rooms impact learner performance and self-motivated engagement in terms of speed, accuracy, and application under pressure?	**Opponent-Based Challenge Module** featuring auto-generated 5-digit access code matchmaking, randomized compound task selection from a predefined pool, and real-time Speed-to-Compound evaluation via Supabase Realtime.

high-speed competitive engagement (*Gamification of Technology-Mediated Training: Not All Competitions Are the Same*, 2016).			

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